

ActInf GuestStream 117.1 ~ Robert Stamps: Active Inference and Artificial

GuestStream | Robert Stamps: Active Inference, Artificial Spin Ice | 1:20:23

All right, hello, welcome. It's August 19th, 2025. We're in active stream, guest stream one 17.1 with Robert Stamps, who will be presenting on control strategies and predictive coding using magnetic noise. Thank you for joining Robert and looking forward to this presentation and then discussion with...

Close the YouTube on your side, Robert.

Okay. Yeah. Oh, shall I just begin now? Okay. Yes.

All right. Well, thank you for this invitation. Maybe I should give you a little bit of a background of how I ended up doing this research, getting interested into this problem because I'm primarily a condensed matter physicist, computational in theory primarily. And my area of specialization has been in magnetic materials and spin systems in general. But I've been doing this for quite a while. And about, I would guess, 25 years ago, I was more or less starting my career.

I was in Australia at the time. And I also came across a paper. It was about the theory of flocking. And I found that very interesting. And that led to one thing and led to another thing. And eventually I found myself co-supervising a PhD student who was working with an expert in psychology on psychophysics. And there's looking, doing experiments and analyzing experiments on primate visual system. And what I found striking was some of the mathematics we were doing. We were heavily influenced by, I think about that time, in fact, just recently, a Rowan Ballard paper had come out. And it was heavily influenced by the structural form of the mathematics we were doing. And it mapped directly back on to some of the things I was thinking about doing with magnetic systems. At that time, people were just starting to develop a fabrication technique which would allow them to make new kinds of magnetic materials. And that's what I'm going to discuss today. And it occurred to me at that time, I remember very clearly that when that technology matured, it seemed to me like it would be an ideal one to try and implement some sort of neuromorphic designs in. And indeed, that is exactly what is panned out. And so I'll tell you a little bit about that and what the state of the field is now. And that's, but that explains more or less how I began my interest in this kind of cross-disciplinary field. And just to put the story back into context, then about three years ago, I had a sabbatical, and I decided to go back to Australia and visit that group and see what had happened. And that's when I came across Friston's papers and active inference. And I actually didn't make head or tail of them. And so I thought, well, I better put it into some context I can understand and try and understand exactly what predictive coding and active inference are about. And that helped me. And that's where this work emerged from, trying to understand that. So let me, let me begin then. So I'll just explain these pictures. This is just a schematic of using nanomagnets with an active inference to control some process and then you feed back. And this is a schematic of the sort of nanoparticle arrays, which are in this control box. This is my idealized spin system. And what this is,

at the very end, I could show you, it's, it's a model, which is a bit more sophisticated, but it actually tries to model properly the properties of real magnetic materials. And the point of it was to demonstrate that, you know, one could implement certain algorithms inside a magnetic context like this. So to begin, let me introduce myself, the people I work with here, focus primarily on studying both theoretically and experimentally nanomagnets. The experimental parts are led by my colleague, Professor von Lierup. And these are students who involved one way or another in some of the works, some of it overlaps with what I'm going to show you today. And these are just illustrations of some of the different systems which other people and which we also have looked at over the years. These, each of these represent a little magnet. But before I explain all of that, let me give you a little bit of an outline. This is a, just an inspiration. It's a wonderful, beautiful picture of a fly larva. And you can see all the neurons here are close packed and they expand out as the fly, as it grows into a fly. But the efficiency and the minimal amount of connections and yet what the fly can do is really quite amazing. And that's a little bit of what trying to discover in all of this. So for the first thing, first thing I'll start with is a little bit of introduction to what I mean by magnetism and nanomagnets and why I think this might be a possible platform, at least for some key aspects of what I think is called predictive coding. And then I'll give some examples, active inference and tracking and some other examples as well. And then finally, when I talked about these real systems and more sophisticated models to demonstrate material specific properties, then I have some micro magnetic examples. That's what they call it. So let me start with a quick overview of magnetism and nanomagnets and some key definitions. So I apologize if you've seen all of this before, but it just gives us a way of defining terms in a common way. So what is a nanomagnet? Well, it just means it's a little piece of magnetic material. This is a real picture. The particles themselves are smaller than a micrometer. There may be 100, 200 nanometers in length and half that size in width. And an example material would be an iron nickel compound. And the shape is what matters here. The shape is such that they tend to align as all the magnetization pointing in one direction or the other direction, but always along this axis. That minimizes the energy of the magnetic atoms. So at lower temperatures, then they form this magnetization and it tends to align along an axis. So it can be fairly well controlled just by the shape. And the lithography has been developing steadily over the last 20 years. So one can even make smaller, well-defined particles. Now, the nice thing about it, by being able to make many of these particles and place them in close proximity to one another, is that these little magnets interact with one another just by stray magnetic fields, the same sort of magnetic fields that holds a magnet to a refrigerator door. But what it does then is, depending on which way the magnet in the magnetic particle is oriented, it always likes to be north-south. So if I draw little arrows to represent which direction the magnetization of the particle is, they always try to order in a way that minimizes the stray field energy, which is represented by these little bar magnets I've drawn schematically over here. And so to be north-south like that means that if they're placing them in this direction, they like to be anti-parallel. That minimizes their energy. Whereas if you place them at a corner like this, then they'll form a corner and try to make a chain. The point is, there is an energy associated with their interaction. They can adjust their orientation to minimize that energy. And that's

sort of the basis starting point for where I think this overlaps with the predictive coding algorithm. So the shape just means they have arrows along one direction and you can place them close together and then they behave like little toy magnets. Now the freedom and flexibility of the lithography, which has developed over the years, is you can place them in all sorts of interesting geometries where these interactions can compete. And the name of that field is artificial spin ice. And it is motivated by a residual entropy that was predicted by Pauli for water ice. The way that the polarization, the distribution between plus and minus charges would order just in regular water ice. It is frustrated. It competes. It cannot find its minimum, minimum energy. And one can engineer that and study the frustration by putting these particles in certain arrangements. So this arrangement is one where there's a two-fold degenerate ground state and it forms net anti-ferromagnet, meaning there's a cancellation of all the magnetic moments. This one, where you just rotate everything a little bit, prefers a ferromagnetic orientation. So there's a net magnetic moment. Others, which can be engineered, are not really in either of these classes. They are instead called topological ordering.

So I'm going to talk about primarily this ferromagnet one, just to illustrate the key ideas here. Before I do that, as I was telling you before my interest in this area, I was thinking that someday people are going to develop these magnetic arrays once I have the technology for neuromorphic application. And indeed, that has been seen in the last five years where there are primarily reservoir computing and algorithms have been implemented. And one of the motivations, in fact, the prime motivation for this is the power efficiency.

So if we think about a conventional semiconductor-based logic, a computer, the information is moved around by charge currents and charge currents lose energy through joule heating, through resistance. So all the heat that's generated by the CPU is just wasted energy for moving charges around. Information in a magnetic system can be encoded in the direction of little arrows I just talked about. So the energy involved in reversing these little arrows can be, in fact, as small as is allowed thermodynamically in principle. It's called the Landauer limit. And this is not just a factor of two or three more efficient. It is orders of magnitude more efficient than what can be done by moving charges around.

So that is one of the prime motivations for trying to implement some of the very computationally heavy activities associated with machine learning. And that's motivated this field. There's some other things, too. You can start thinking about modifying individual elements and states. Next horizon is actually being able to do this in three dimensions.

And people are making strides toward that. So it's an interesting area and it has the promise, or at least the potential, of introducing a non-semiconductor and conventional platform, which is particularly power efficient for conventional and non-conventional computing.

So, to make use of all that, there's some basic concepts. So I'll begin with this design here of a spin system. Now, it prefers to align in its lowest energy configuration such that there is, if you add up the little arrows, there will be a net magnetic moment, meaning there will be a net number of arrows which don't cancel out.

These cancel out in that direction, but this group of four, those add.

And there's four different ways to do that. It could be these two adding and these canceling.

And what we care about is the average direction of the arrow, which is the average magnetization.

So I define a magnetization then as an average, and it will be a thermal average, where these little sigmas represent the individual spins making up the magnetization.

And this can be averaged over the entire material.

And so if you do that, you would measure then the average magnetization, the way these superimpose with one another, so that you have a magnetization curve, which would look as a function of temperature.

They all, for a ferromagnet, like to lie in the same direction, or nearly lie as you increase temperature.

But at some critical temperature, spins will start reversing and point in another direction, and that reduces the magnetization.

And the rate at which this happens, we call susceptibility.

And statistically, that's actually a correlation function.

Those two things are important because those are the two things which I will try to convince you overlap directly into a predictive coding scheme and just enable it then the magnetization to act as a platform for implementing this sort of thing.

So the next bit that's important is to recognize how the magnetization evolves over time, the processes involved.

So if I consider a whole lot of these little magnets, whatever geometry it is in, I can adjust the orientation with a magnetic field, which I might apply externally.

So I might generate a magnetic field either through another magnet or a coil or something like that, and apply it to this array.

And that, if it's strong enough, it will tend to align all the magnetizations as much as they can in the direction of that field.

Now, if I reduce that field, and I have a finite temperature, there will be a reduction in the number of spins or elements which like to point in their original direction.

And some of them will start to favor alignment with the field in its new direction.

And so there might be a region which is pointed oppositely the others.

So you can see there's a net reduction in the magnetization.

And as I increase the magnetic field in this new direction, this region grows at the expense of the other region until when I get to a large enough field, everything is pointing in the opposite direction.

The point I want to make here is this is all happening in time, and it can be observed in real time.

So here's an example of an experiment that was done on such a system where they start the magnetization in one direction, then apply a small field to give it another direction, and then watch what happens as it evolves.

And so microscopically what happens is all the spins were oriented one way, and then one starts to reverse, and that causes or makes it more probable that another spin nearby will reverse, which makes it more probable that yet another spin will reverse until they all reverse.

This all happens in real time, and there's a statistical probability of each of these reversals.

So that time that it occurs in always changes a little bit, but it is a stochastic process, and each reversal is

sampling the configuration space, but it's always looking for a way to minimize its energy.

So it's always trying to relax into a lowest energy configuration.

So that sets me now to where I can talk about the overlap with predictive coding.

Everybody here, I think, knows much more about this than I.

I've been struggling to understand the logic behind it, but I think I've got a better idea now.

If we think about two variables associated by, I think of it this way, if I'm looking at the visual system, which I'm a little bit more familiar with,

I throw an object in the air, my eyes are following it, but they're doing it in a semi-random way.

They're saccadic eye motion, and they tend to, where am I going to look in the next instant is going to depend on what I've perceived in the previous instance.

And so there's a lot of information there, there's velocity, position, all of that, but I'm going to look in an area where it's probably more probable to find the object in the next instant.

So there's two things going on. I have a perception, and I have some expectation or belief of where it's going to be.

And what I'm interested in, or what my neurons are interested in, is a probability distribution relating these two quantities.

Now, and that's where the whole theory begins. So I've got some sort of input data, and I've got some sort of belief state of my neurons.

And I'm going to say there's some sort of mapping, and it's not a perfect one-to-one, it's a stochastic, it's a certain randomness on both sides of this.

And I'll say that there's a nonlinear function mapping it, or it can be nonlinear, mapping, say, my perception to a belief with a bit of random noise as well.

Or maybe my next belief, the transition to my next belief is also going to be a nonlinear mapping, or linear, it can be linearized as well.

But it's some sort of mapping, some function. I don't necessarily know what that function is.

And moreover, this can be in a hierarchical system. So like the original Raoul Ballard paper, where a lot of this was described in terms of vision, is a layered system, where there's an image input from the back of the eye.

And then that feeds forward into regions of the brain, there feed forward differences between the expectation, what was perceived, and then a prediction of which way maybe the eye ought to move next, or it would involve that.

And it would keep going up the ladder and back and forth.

But I think that the main idea is that each layer tries to minimize locally its error function, and match up the predictions with the perceptions at each layer.

So this is a little bit different, I think, than a lot of global-oriented machine learning algorithms that are maybe a bit more familiar with.

One other thing, too, when I gave my example about tracking an object, there's a lot of information being gathered in each time interval.

And that includes position, but it also includes velocity and accelerations, and can in principle be any order of dynamics.

And so in addition to having a hierarchical layering, one can also envision a kind of a lateral hierarchy, if you want.

And so I call that dynamic ordering.

And that is, I think, the idea that Friston introduced back in the late two, around 2008 and so on.

And what the idea that evolves from all of that is I now have new indices that complicate my states and my perceptions.

They are indexed by layer and by dynamic order.

But the time evolution of my beliefs then, with this dynamic information, also introduce a term over here, but they're governed primarily by this relaxation of some energy function.

And that tends to, that's the free energy idea that you're trying to find an extremal value.

And it's this gradient which is most interesting for me, for what guides this, because that's where I can map that onto the magnetic system most easily.

And I put a little note here that this is a Langevin-type equation, but it can be also derived in a physics context,

where most physicists would be familiar with the Fokker-Planck equation for the distribution, the time evolution of a joint distribution function.

And these equations emerge naturally from that.

Here's how we do it with predictive coding then and magnetization.

So I've got my state variables.

You might call them the hidden states and the more visible states.

They're indexed by layer in the hierarchy, but also by dynamic order.

And I'm going to say that I'm going to translate this statement of a nonlinear mapping between the visible and hidden states through these G and F functions.

I'm going to translate them into measurements on a magnetic system.

And the magnetic system is a magnetization, something which responds where I put a field on it.

That field is a function of these variables, these state variables, and the magnetization is what responds or how the system responds to that magnetization.

As I said earlier, the system is always trying to find a way to minimize its mechanical energy and its entropy, maximize its entropy,

subject to whatever that constraining field is.

So if I can sample that, that's why I put these angular brackets, it's a stochastic sampling,

then I get a measure of the system trying to relax, as it must, to a more energetically favorable configuration.

And I just interpret that relaxation as an instantaneous value of the visible and hidden states.

So this is all done at some finite temperature T .

And the trick then is to figure out what this field is and then make a measurement of the magnetization.

And so my Langevin equation then with this gradient of this error term or energy translates into a transition rate for my state variables,

here the belief, subject to this term which is effectively a force guiding the magnetization toward a lower energy state.

So the terms that appear just for some details and how the mapping occurs, the field, which is the all important thing that's acting on the magnetization, is a gradient of some energy function.

And that energy function is the lowest order expression for that is the difference between this nonlinear mapping and the average of my state,

whether I'm at whatever layer I'm in and at whatever order I'm in.

And that defines my effective field.

The terms which appear then, there's a sum parameter, a precision,

which in my magnetic system goes like an inverse temperature,

and magnetization, which is my F-function,

and the derivative of that F-function, which is my susceptibility.

So all these two things they can get from a measurement.

I do it computationally, but they can be done through measurements.

We do that for data recording, for example, magnetic data recording all the time.

And this is the quantity then that tells me how my state variables will evolve in time.

So the algorithm is really actually quite simple.

At each time step, I sample my magnetization on some magnetic array,

where I've subjected it to a field, which is a function of whatever the previous values for the state variables were.

Then having measured that magnetization, I get new values for my state variables,

and I calculate fields, and I repeat the process.

So it's nothing more than that.

It's the same thing we used to do with magnetic recording, still do do with magnetic recording.

You have information on a magnetic material, you read it, but you can also write it.

You just repeat it.

So it's really quite simple, and it's an old, fairly well-established technology.

What's different is doing it on micrometer size patterns.

Well, that's not different either.

Here's an example.

So here's a three-layer network with three dynamic nodes,

which would correspond to three different magnetic arrays,

and I would apply a field separately to each one of these arrays in my updates,

and then I just make a measurement of the magnetization from each of these nine arrays,

feed it back in to get nine more fields, and just keep iterating.

That's my implementation of the Longinian equations that govern the predictive coding.

And that's how it works.

It really is kind of simple.

Here's an example.

So I've done one which is fairly recent, actually.

It's not dynamic.

It's just the layering.

I modeled it on the Raoul Ballard paper,
but it's a simple one-dimensional version.

So I have an image with one dimension.

It's just a sinusoid, so I got a feature in it.

And I don't have very many nodes.

I think this was like 30 or something like that.

And then I reduce it to my first active layer, μ_1 , which I think had something like 18 nodes.

And then I reduce it down to six nodes in μ_2 and a bit more feature extraction with μ_3 .

And then for μ_3 , I can do a reconstruction.

Now, over here, μ_1 and μ_3 don't look very good.

That's because I'm doing it at high temperature.

My original image looks like that, and the reconstruction looks like that.

When I lower the temperature, because remember, you know, I get a critical point where the magnetization disappears.

When I lower the temperature, well, quite a different picture.

Here's my input data.

Here's my first layer.

It's learned it pretty well.

Here's the second layer, which is extracting some features learned from this and more refinement at this layer.

Then I reconstruct from this layer the full image, and it matches pretty well.

But that's at low temperature.

So this just illustrates, I think, where the magnetic system, you can actually see what it is actually doing something.

And it's through this nonlinear function.

Another example.

This is now one which includes the dynamics.

So it's a tracking example.

And I was inspired that there was a thesis written by Millage.

I think it is.

It's quite nice thesis.

And in this chapter three, he did something like this.

I thought that was quite nice.

So I implemented it here.

And this is a sinusoid.

And I give into it, because I've got three dynamic orders as well as three hierarchical orders.
So I have a velocity and acceleration and an acceleration to the acceleration.
And that input data then is represented here with my ϕ .
And then this is the layer one.
And it's following the sinusoid quite nicely.
This is the layer two, which is using a visible layer from here.
By the way, I'm following a paper.
The formalism I'm using is by Buckley.
So I have that hierarchy that they've used.
So this is the visible and their hidden notation.
These are errors on that.
And it's following quite nicely.
So I was curious how this, what's being learned, if anything.
So I tried a non-periodic input.
So I would just, at each position, choose randomly a velocity.
So sometimes go up or down.
It's a one-dimensional example.
And you see, it does pretty well following even that random choices in velocity.
So I ran it for some time.
And then I turned off the signal.
So I turned off my input here.
And as you can see, the hidden states, however, they keep going on.
Looks like they learned a trajectory from the most recent steps.
And so it appears that it did learn to predict which way the system was going to go.
And yet you can see here, it makes very quick corrections.
But here there's no input data.
And it will eventually come back down to that.
And this is going, a lot of stuff is being filtered at the higher layers.
They respond a bit more quickly than the lowest layer.
And that's all consistent too with the observations that different timescales are happening at different hierarchical layers.
Active inference, which I haven't done yet, but it's here.
This is this example.
So that's what I showed you on the very first slide.
Nanomagnet control, feedback to a process, and then information coming back to the nanomagnets.
And again, this is just repeating what was done in the Buckley paper.
The active inference part is this A.
So we have the, we've associated then some change in the process, which appears then as a new function in

this E function.

And this appears in the perception part.

So I look at how that is changing and adjust whatever this A signal is based on what is believed by the nanomagnets.

And the target here is a linearized F function, but I'm still using an average over that linear F function of, from my, simulating it with my thermal magnets.

So introduced into that function is a target value, TD.

And what that means is the system is going to try and align its perceptions through A with its belief that it ought to be at a temperature TD.

And this function that is going on in the process is unknown to the system.

So it has to, it can't know the function.

It just has to adjust its position with μ in such a way as to get the data coming back into the nanomagnet to give it a value of TD.

So its target here is the green line.

And here is the measurement is the orange.

And the belief is the blue.

And you see it starts off over here, far away from the target.

And then it oscillates and exponentially decays down until it hits the target.

And this is the other bit of information that's coming in.

That's the temperature.

This is the rate of bit change of the temperature with respect to whatever this value X is that it's sending to A.

And so it's got these two bits of input, which is just the position and the velocity.

Whoops.

Went backwards.

And then this is the error function.

And I found it actually useful to see this as a trajectory.

So that if I look at the belief in blue, which is really what my spins are doing, and the perception that's coming back in, which is my ϕ .

And they're starting off here, and then the perception is slowly coming around and oscillating into its target value here.

And you see that the blue, which is the stochastic magnets, they're oscillating as well, but doing quite a bit more jumping around.

But they also then find their center of gravity at the target point.

This is just a note that rather than doing the simulation, I can actually capture most of what's going on here with a simpler approximation, which is quite common in magnetic systems.

It's a brilliant function, which allows me to introduce temperature.

And you see, it gives me the same qualitative features that I had with the actual magnetic system.

This is another thing, which is that system below, before, was just for three dynamical orders and one layer, one hierarchical layer.

There are parameters which matter, these precisions on two sets of terms which appear in my effective energy.

I can get rid of the oscillations by adjusting these parameters.

So what I did was I basically built an active inference acting on the acting inference, if you will.

So I built another network, again with magnetic spins.

And their job, however, was to measure the energy cost of the first layer, which is interacting with the process, over time.

So they were running much slower.

And then try to minimize that energy by feeding back to these parameters, the λ_z and λ_ω .

And so this was the first iteration.

It just got started.

But now I collected data.

And then there's feedback from this active inference acting on the acting inference, which got fewer oscillations.

And as I repeat this through this learning exercise, you see it gets quite good near the end after about eight or nine repeats.

And it just goes smoothly to the target.

So it's adjusted these parameters relative to the starting point to take it exactly to the target without a lot of searching around.

The last thing I'll mention is this.

We're done with fairly simple models.

That is, simple spins, real magnets that people can make are not that simple.

Although I did demonstrate you can use a fairly straightforward mean field theory to get the essential behaviors.

But that's a far cry from actually making a device.

So there are tools for trying to simulate better, more accurately, more realistically, magnetic nanoparticles.

So I applied those tools.

And they can be done at finite temperatures.

It's just a different algorithm.

And it's called micromagnetics.

And I can put in parameters which are measured and known for materials, iron, nickel being one of the compounds and various variants of those compounds.

But there are things that people can actually make lithographically into the kind of designs which I was showing.

So I chose one with 16 spins.

And in this kind of a geometry.

So it's much simpler in a system than what I was showing before.

And I did this tracking exercise, but now call it a PID controller.

This is what it would look like electronically.

What I did was the same thing, except it would be done with 16 spins.

And it did the same sort of targeting adjustment and oscillates around here, where I'm now measuring directly the magnetization through my micromagnetic simulations and then adjusting those for the applied field, which then is feeding back into my imaginary thermostat over here.

Another application of the 16 spin system.

If I, say, feed in an input function, which is a sinusoid and a cosine for my two dynamical orders, I can move the system of my 16 spins such that from one initial condition to another, which I specify, will correspond to an average magnetization.

And that average magnetization is nothing more than the average number of spins pointing to the right or up and down.

And I can pick then a set of states by picking my starting points and my final points.

I can target a little subset of states where the magnets, specific magnets, are pointing in one direction or the other.

So I can pick examples of this has a certain magnetization, this has a certain magnetization, corresponding to the alignment of the spins.

I can choose those states with some precision just by picking my initial and final conditions.

The final one is just an example that what is a computationally hard problem in the field, which is if you have a real nanoparticle, some finite size with a lot of different kinds of more complex interactions.

How do you reverse it in a very, very short time?

And this algorithm works rather nicely.

It reverses a magnetization from pointing up here to pointing down there by applying a pulsed, shaped pulsed field in the right directions, moving the field as well, which is a really difficult problem.

You can't solve that very easily using any of the tools that people have at the moment.

And that's it.

That's my short summary.

I think maybe nanomagnets and I think a broad class of systems probably provide natural platforms for predictive coding by making this sort of mapping onto the magnetization or whatever the order parameter is and the susceptibility,

which are all temperature dependent.

They're also stochastic processes.

They are minimizing a physical free energy and you can just map the two onto this apparently and constrain the system to evolve along a trajectory, which we would recognize then in an active inference example.

I'm also particularly interested in systems which are complex and like glasses and spin glasses.

Some of these display all sorts of various odd and not very well understood dynamics.

And I'm wondering about whether or not there's some avenues into those studies by employing the sort of a methodology, if you will.

That's it.

That's it.

So.

Thank you, Robert.

That was awesome.

Anyone watching live, please write any questions.

First, Vladimir, please feel free to give any opening remarks and ask any questions you have first.

Yeah, this is very interesting.

Presentations and I have several questions for this.

So what your system, what I see that it works and it can do calculations about what you said, predictive coding is mostly doing computation with the matter.

And in this system, I think there's a main part that it is based on the mean field.

Yes.

So that you are using Langevin equations and this is going to the free energy minima.

And so to go to the minima and this system just learn the path.

And that's what you have this gradients towards the minimum.

And so this is the basis of the system.

The why it can be predictable and then how you can calculate.

Yes, this is correct.

Yeah.

Yeah.

I know that's exactly it.

That's why I think it's rather general.

It can be taken outside of magnetic systems as well.

Yeah.

Yeah.

So this is quite general.

So you are at equilibrium.

So you have several local minima and you can even make those changes between the minima.

Yes.

So for example, if you have a system with several minima and the kinetic barriers, so that's, can you also incorporate this?

Well, that's what I would like.

That's kind of what motivated all of it.

As I mentioned at the end about glasses, which would be exactly that sort of system.

And it's not really at equilibrium.

And that's what makes it interesting, right?

Because this evolution, the Langevin equation evolution, especially with the dynamic orders, is tracing a path through phase space.

And this is introducing constraints on that path.

So I'm wondering, what I'd like to know is, does the landscape, the energy landscape, is this a way of studying it?

What role does it play?

Yeah.

Yes.

So that's basically, if you have several local minima and switches, it may be quite interesting for computation.

So if there are, those kinetic barriers are too high, then you are too stable, it will be not possible.

And if it's too low, it's also unstable.

So this kind of things.

And another thing that would be also interesting to see how you can do for this non-equilibrium systems, which are not based on this relaxation or this Langevin, but maybe going towards non-deazol.

So I don't know if it's interesting to explore this direction.

It's completely non-equilibrium.

I could say, glasses and jumpings.

Yeah, that's exactly what I would like to do.

So far, I haven't.

I mean, one of the reasons I chose a spin the ice as a model system is you can engineer the landscape to some extent.

Yes.

You can have multiple, multiple energy states.

And I have, I've explored it numerically, primarily trying to see the consequences.

I found some oddities, some things, but I haven't, haven't found, I haven't made sense of it yet, honestly.

And have you tried it to, because this system is a small size.

So this is, you can have a very big number of those particles or those spins.

And these big numbers may lead to some different scale, so that you can do not only just explore at minimum, but you can maybe use it as a, something called a completely different scale.

So if you, if you have access to such a big number of, uh, interacting particles, so you have a, like, uh, like an Avogadro, uh, they lead to some new phenomena.

I wonder that too.

I mean, computationally, right.

It starts to become very difficult.

Um, and, and, uh, we, we, we're, we're kind of moving in that direction, trying to, to be able to treat larger systems.

Uh, but it, it, it is, it starts to become very demanding.

Uh, for what?

To control it?

Or, to calculate, to simulate it over, over sufficient long time scales.

Yeah.

But if you leave it to evolve by itself.

So you, because you want to always like, uh, external control, but you, what it does.

Yes.

But if you just switch by itself, so just to self evolve.

Well, that's it.

I'm not, yeah, I'm not sure I'm following.

Uh, how would you, what would you look at?

What would you look at?

How would you set up that problem?

Uh, so for example, you may have, uh, uh, or you measure the properties, for example, monetization, susceptibility of the whole system.

So this is there, but you don't provide the program what you, what, what it should calculate, but if the system itself organized into some states by itself.

Uh, let's see.

So, for example, I played around a little bit with this idea.

Um, if I have, I mean, instead of a large system, maybe I have many smaller systems, which I run simultaneously on the same problem, but I let each one follow its own path.

And so for example, I think I was playing this when I showed you an example of the tracking where I'm, um, using a active inference to adjust my parameters to optimize parameters for the tracking.

Um, if I have several systems simultaneously doing this and letting them adjust their own parameters, so to speak,

then I have, uh, uh, I think I was trying to think about which one, one of them is going to give me, uh, uh, uh, uh, state,

which is the one I want to feed back to my process because I can't feed back all of them.

So I've got 2,000 different pro, uh, uh, uh, uh, systems, which are all trying to predict an outcome, but only one of those is going to actually feed back to my process.

Uh, which one do I choose?

So I, I can make a, uh, uh, create a probability of which one, maybe the one that is closest, that predicts closest to the target value has a highest probability.

So I could choose which system to follow based on that probability.

So I'm sort of sampling between all my, my different, uh, uh, uh, systems.

Is that sort of what you're thinking of in a way?

Yes.

But if you consider that interaction, not calculation in a, uh, in a normal way that you provide a program or you design all and control external control, but, uh, the material, what you have, interact with the, uh, environment and adjust the parameters without, uh, even finding these, uh, instructions. I'll ask a question from the live chat or I, I, I think I hear you on that Vladimir.

I'm imagining one scenario where just stimuli are presented to a material and it could just be like sound vibrating sand and the sand pile never at, or maybe it's like an ultimate unsupervised learning.

Yeah.

But how much of this is the design for situation specific topologies like circuit design?

I, and it's, is it more like an ASIC and it's more restricted to a certain kind of inference?

Or how, how do you think about what functions it can perform?

Or is it a general computer, like a processor, even if you could only do it very slowly?

Like you play game of life, that kind of stuff.

If it's just like a smaller hard drive with a spatialization element.

I re maybe this is relevant.

I'm not sure.

There's this, this thesis by millage, um, in this chapter, he, he did an image learning.

It was like learning, um, the MNIST digits, but he also had an interesting comment about, uh, hallucinating.

If he gave it a partial or, uh, really a minimal amount of input, the system came back and filled in missing gaps in a, a way, which was a bit surprising.

So it was, was like the system was, was self organizing into a response to something it couldn't categorize very well.

I thought that was intriguing.

I didn't really know what, what that was saying, but it seemed interesting.

One.

Yeah.

That, that makes me think of a free energy minimum being possibly a local or a global minima, but still being very far off from like a, some other, some other measure.

I don't know.

Another one example, which I don't understand, but I, I found very interesting.

There were these papers, a few papers by, by Friston on, uh, uh, a duet for one.

If, if I don't know if you know these papers with, uh, uh, it was about birdsong.

And I, I think in one of them, if I remember correctly, yeah, it was, it was almost as if there was one system, uh, uh, uh, but then it was, it was modeling it as if it were also two systems.

So we have one, which is singing and then it stops and the other one picked up the song, uh, and reproduced a lot of it.

That, that almost seems a little like that as well.

It's kind of in that category.

It's learned that it's following it, but it's responding then because it's, it's, it's, it's up in this higher layers and comes back down.

I'll ask a question from live chat.

Bert wrote, I did not understand the step from the predictive processing feedback loop to active inference.

Can you explain how to add in action?

What are the affordances?

This one, I think.

Um, so what I have in this case, do I have one?

I have in this case, do I have all the equations?

Um, so there, there are two streams of input to the nanomagnets.

One is the temperature and one is the rate of change of the temperature, uh, when you adjust a parameter.

Um, and the parameter, this X is essentially the, the rate of change of the perception with respect to whatever value of A is in, in terms of the magnets.

So it's adjusting A in a way that is proportional to, this turns out to be an error term, uh, on, on, uh, the prediction and what is measured.

And then this coefficient just measures the, the response essentially, uh, that this system is going to, to react with.

Uh, so it's, I've got, I've got a single layer.

And then there's the system down here, which is providing information.

So it's a feed, feed forward, but then this is a feedback is what the A value is.

So it's trying to minimize a discrepancy between the belief and the perception by adjusting some parameter X .

And all it knows is it changes this parameter X and there's a response, uh, uh, that leads to a new response from the process.

So if it changes it and it gets worse, then the signs on that term change.

And so it pushes it in the other direction.

I don't know if that, does that answer the question?

So you, you mentioned the frustration and the study of frustration.

So it's kind of like there's one alignment where it's a minimum, a global or local minima, which can be designed to be a, or modeled as a convex optimization, at least locally.

And then that, the system could be like the magnet is pointing north and the compass is aligned.

And so then that's a static state or the magnetic field could be rotating or there could be other forces on the compass needle.

And then the overall like tension or frustration is basically like the, the relative malfitness.

So that's the downhill tension reducing force.

Mm-hmm.

That at least locally, the, the compass is going to reduce its, the gradient is going to go one way or it's exactly at a flat point.

And then it just is static unless there's some other output.

And then to the physical part.

So then maybe to end the first question about what the affordances are in the lab, what kinds of physical measurements and interventions can you do?

Well, for the magnetic system, the kind of measurement you would do is measure the magnetization.

A simple way to do that is if you've got a coil of wire and you move the magnet by it, the field which passes through the wire induces a current and that can be measured.

So that, that, that, that's sort of the basis.

There's other ways to measure magnetization, but that, that's kind of a simple, straightforward one.

Is that, that you mean the mechanics of measuring, of measuring the magnetic particles or is this a different question?

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by that.

Vladimir, do you want to ask another question?

No, I think it was quite clear that this is, I think, this is a very interesting study and in the materials it is not yet, this direction is very new.

I think it's not yet how to compute with a matter, it's a quite new direction and that's one of the examples how it can be realized experimentally.

Because in many discussions in the publications, there is a lot of theoretical concepts, for example, this active inference and other theoretical concepts, how it should be done and how it can be realized.

Experimentally, very few examples which are really implemented.

And this is one of those examples.

I think this is the approach, this is a demonstration among the first.

I have a question for you also, Vladimir, if that's alright.

I mean, I think you work with, I think I saw through some of your publications work with nanoparticles.

Do you work with materials that have some sort of polarization or something analogous to... nanoparticles, which is a very interesting question.

There was mostly a mean field, yes, it's one of the directions that's also how to go to the equilibrium and how to minimize the free energy in general.

So this is about, for example, for interaction with the membranes and biological systems.

There is also quite a big study.

So this goes in this direction.

And now what we are building the network for materials, with intelligent materials.

And here, in one of the directions, what we were trying to do is to build the discovery engine.

And also with Danielle, it's involved in this, how to also for this field of materials, how to find concepts.

For example, how to make those materials intelligent and what are the phenomenological concepts.

So on a very high level.

So what do you need?

Memory, you need computation, you need what is the time scale, so the system should change.

And what is the short term, what is the long term.

And this depends on the relaxation time.

So kind of this, because it doesn't matter what is the system, but you can already see if, how it can behave.

Because this, as you see, this is based on thermal energy, minimization of the energy.

And there are very fundamental concepts which are driving, driven them.

And the realization, it's already can be captured from one system to another system.

And you can even map it.

And see, this is your system of magnets, what you see, it's quite general.

The mechanism is quite general, can be applied to another system.

So this basically, the idea of this work is how to make those connections between different fields or different materials.

And if you, and the major problem there is that you don't have reliable experimental systems which are demonstrating it.

Because there is a lot of theoretical concepts how it can be done.

But you need to have this demonstratable, falsifiable results.

And this is, I think this is how you show it.

It can be used for another systems.

This is kind of, it can be mapped.

Thanks, Slade.

I'll add like one kind of connection there that, arising from your talk is, how do we think about the aspects or the capacities or, or reliability?

What to even measure?

How to have a useful engineering and in silico and in material flexible?

So that you can study different approximations and more field-like models and more point-like models.

So that you can do normal statistics on it and kind of port or have a good assembly language type connection.

Kind of between like, like the analogy between processor instructions and the digital compute.

So then it's like, well, what is the approach to map different materials and systems to their functions?

And then that's where these cognitive ontologies, like precision, attention, belief, updating, learning, all of those kinds of cognitive ontological terms,

become mappable from materials.

And then that's where with that sort of team and work, it was interesting to look at the method sections and look at the parameter spaces of where methods have or haven't been done.

Like which grid sizes have been done.

So that's where it's just a really high dimensional space, but possibly could be navigated interestingly to understand where, where there might be, where is something like a material that emulates a given function.

Where is that, like, like, what is the status of being able to make that?

And then understand that within a structured parameterized experimental space to prioritize different experiments and perform meta-analysis to understand how, like, just all, all these different pieces.

So, yeah, just to add a little, from what I've seen from that project and where this sort of talk takes me.

Hmm.

Well, I think I'd be very excited if someone would actually try to test this in one form or another, this idea.

I think, yes, you could maybe, I chose magnets just because I'm familiar with them.

But, I mean, other platforms might be easier technically to implement some example.

I think the PID controller is a nice one because it's close to an application.

It's used in an enormous number of things, industrial control to cruise control on your car, for example.

Or just a thermostat for your home.

I mean, some practical device demonstration might be motivational, in a sense.

Magnets may not be the best way to do it.

Maybe there are other systems where you could envision something analogous, liquid crystals, or something which has a non-trivial response in ordering,

and yet subject to thermal processes.

I know that even in some convention, memory storage technology, for example, is something which over 10 or 15 years has developed slowly,

but it's kind of in that direction, material platform, where you have something which has a transition function, which would play the role, say, like the F function, for example.

People do look in that direction as well.

And economics is the motivator there is to try to find more power of efficient replacements for what one does with transistors.

But I mean, in trying to find implications for something like predictive coding, you've got lots of examples from nature.

You can get devices that could do something similar.

That's all known.

Is there a software toolkit or a production possibility?

Like how would we....

What kind of modeling can others engage with from this work, or production of these objects? experimentally.

I'm not sure I'm following that.

Yeah, like are they synthesizable by you?

Yes.

Or how do they get synthesized or designed?

Or what open source or technical documentation exists to design the in-silico models or to make them physically?

Well, I mean, in my experience anyway, if you have an idea and something that could be measured is to convince someone who has the laboratory able to do it to invest a few years of their time and graduate students into making it happen.

So that's a little bit what I tried to do with this paper is demonstrate that, yeah, there would be a possibility of making these things using existing technologies.

It could be done and someone may try it.

But that would be just one example out of, I think, a new big possibilities, a big problem, possibility space.

It's great ideas of other systems which might be easier to realize.

Well, I think in this one, that's one of the reasons you chose it for this, we have laboratories here can make nanoparticles and they do study spin glasses.

We may be able to do something in that direction.

Yeah, it's very fascinating.

Connor Hines and others have worked on the spin glass system a little bit as well.

So finding that sort of isomorphism between a generative model, which can be emulated with distributions and maybe on batch data that works fine.

But then being able to have those calculations, which is a very small set of them possibly, or even just one, like gradient minimization methods alone, and have aspects of that computation, just like sometimes we amortize from the matrix to the neural network, and then go one layer into basically outsourcing some of what appears to be the signal processing and action selection like features of systems and have that based upon the materials properties.

And then co-design, well then what symbolic aspects complement the materials properties that might handle what otherwise seems to be variants that needs to be soaked up by an algorithm?

Within just like the big data processing framework.

I think the spin glass, I think you mentioned the paper on spin glasses and active inference, Heinz I think it was.

Yeah, that's an interesting one because one wonders is there a way of approaching that sort of a problem from a different perspective, which is less coming up with a device or an algorithm, but it's just a way of thinking about things, which I do wonder about.

Maybe a direction to look would be in active systems, which is, I think, becoming, I mentioned this paper about bird flocking years and years ago, but that kept up.

And I think the modern name for this is active systems, falls under that rubric probably, where you think about particles which have almost like an attached motor, and then how do they, what's the hydrodynamics of such a system.

But I do wonder then if this is this way of thinking, way of modeling, wouldn't be quite a natural approach in some aspects of that kind of a more complex system.

The question for me though is how to use this analysis in a way which is illuminating.

I'm not sure I know yet.

I mean, I did get some insight in the exercise here.

I didn't mention it, I didn't talk about it, but I found it useful to take these equations that govern this PID.

It's a set of three Langevin coupled equations.

And if I look at it from a physics point of view as a set of equations which are not stochastic, but let's suppose we just take those differential equations and study their spectral response.

Then, if I, by choosing my a function as simply as possible, one can show that the system of equations is oscillatory with an exponential decay.

And that's not very surprising.

It's just an oscillating system that responds in a well-defined way.

What is surprising is that it will go asymptotically toward the target, whatever you set the target at.

But when you introduce noise into the system, it finds it much more quickly.

And that helped me understand how these equations are working.

They're not mysterious.

They're just a mechanical response, if you will.

But it's introducing the noise which actually makes it work better and faster.

That surprised me.

So, that was maybe an insight I can take to be looking at other systems which are relaxation dynamics, which are non-trivial.

So, the noise matters quite a lot.

The magnitude and how it's introduced.

That is interesting.

So, that's a direction I wonder about.

And the same with your question, Vladimir, about a system that might have a rather complex landscape.

I think that's interesting as well in terms of how that evolves on a longer time scale.

And this is also related, for example, with your example with the noise.

The stability of those systems.

Because it is possible to design theoretically some system, but experimentally it will not be stable.

And this is the major problem that how to control it, that it will not fall apart.

So, the noise may destroy the properties.

And how do you control it?

Yeah.

Like you showed that above a certain temperature, the function, so noise or temperature, variance of one kind or another.

It's just, there's the right amount.

So, that's conveniently framed as precision or cognitive attention.

And then, I think the biggest theoretical mapping is that there's no interpolation of a reward secondary function.

You're able to directly have a physical analogy, at least where the analytic form of the free energy functional in silica can be designed to reflect what was already transferred from the physics or inspired from the thermo and the information.

So, the theory mapping to the materials, like Vladimir pointed out, is really an exciting edge that your work is pushing along.

So, it's awesome to hear.

Especially to find the parameters.

Yeah.

There is a certain range of parameters when the system is behaving like it can compute and it is stable and we can behave.

And that's the same range of parameters.

But those parameters are not obvious.

How to find this window, all these parameters, temperatures and all the distances and everything that you have, these control parameters, they should be in a narrow range that it has sense.

And this is very difficult to find.

And that kind of connects to the active inference for active inference that you talked about, like intelligent selection of materials properties.

So, thank you again for the work and for sharing.

Hope that people are interested and learn more.

Hope that people are interested in learning more and you and your colleagues are welcome to share more.

Well, thank you very much.

It was nice to learn about the work that you're doing and the discussion.

So, thank you.

Thank you.

Thank you.

Thank you.

Thank you.